

Diagram illustrating the Tucker decomposition of a 3D tensor T into a core tensor G and three factor matrices A , B , and C .

The tensor T is represented by a purple cube and is labeled $T \in \mathbb{R}^{n \times m \times p}$.

The decomposition is shown as a sum of terms:

$$T = \sum_{r=1}^R a_r \otimes b_r \otimes c_r$$

Each term in the sum consists of three vectors:

- A blue vertical vector labeled a_r .
- An orange horizontal vector labeled b_r .
- A green diagonal vector labeled c_r .

The vectors are connected by plus signs, indicating the summation of the outer products.